3. Terminology

Before diving into the system design, it is important to clearly define the key entities and financial concepts involved in the payment flow.

1. Merchant's Customer

The **end user** who initiates a payment to purchase goods or services from a merchant.

- Example: A customer buying a product from an e-commerce website
- They provide payment details (card, UPI, etc.) to complete the transaction

2. Merchant

The **business or platform** that sells goods or services and integrates with the payment gateway.

- Example: Amazon, a SaaS platform, or an online store
- They are the direct clients of Stripe (or any payment gateway)

3. Stripe (Payment Gateway / Processor)

In this system, **Stripe represents us**.

- Acts as both:
  - Payment Gateway (accepting and routing payment requests)
  - Payment Processor (handling authorization and settlement workflows)
- Responsible for securely moving money between customer and merchant
- Ensures compliance, security, and integration with financial institutions

4. Acquiring Bank

The **bank that holds Stripe's account** and receives funds on behalf of merchants.

- Acts as the receiving side in the payment flow
- Settles funds from card networks into Stripe's internal accounts
- Later enables payout to merchant bank accounts

5. Card Network

A **financial communication network** that routes payment information between banks.

- Examples: Visa, Mastercard, American Express
- Responsible for:
  - Authorizing transactions
  - Transferring payment instructions between issuing and acquiring banks
- Does not directly hold money

6. Issuing Bank

The **customer's bank**, which issued the payment card.

- Responsible for:
  - Approving or declining transactions
  - Deducting funds from customer's account
- Example: HDFC Bank issuing a Visa card to a customer

7. Payout

The **process of transferring funds from Stripe to the merchant's bank account**.

- Happens after successful payment settlement
- May occur:
  - Instantly (instant payouts)
  - On a scheduled basis (daily/weekly batch payouts)

8. Ledger

A **core financial system of record** that tracks all monetary movements in a double-entry format.

Each transaction is recorded as:

- A **debit entry**
- A **credit entry**

Example:

- Merchant account: +$100
- Stripe account: -$100

This ensures:

- Every transaction is balanced
- Total sum across all entries is always zero
- Full auditability and traceability

Key Properties:

- Immutable (records are never deleted, only appended)
- Append-only system
- Required for legal and financial compliance
- Source of truth for all balances and settlements